XRPL Validator — Daily Engineering Report · 2026-09-18

Period: 2026-09-18 00:00 to 24:00 US Eastern (2026-09-18 04:00 to 2026-09-19 04:00 UTC); ledgers #107,061,365 to #107,083,733. Every count from the ledger and the validator's logs stops at #107,083,733. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** 14 engine receipts recorded (5 fixed, 0 explained, 9 without a recorded fix).
- **Changes:** 2 deployments (cycles 141, 143), 57 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

Ledgers with a receipt in the period (#107,061,365–#107,083,733), by step:

Step	Ledgers with a receipt
Step 1 · native engine	14
Step 2 · records	0
Step 3 · order	0

Step	Ledgers with a receipt
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#107,063,938	Step 1 · native engine	1 transaction result(s) differed from rippled's; 8 change(s) of ours the network did not make	fixed: commit 67b0d6d, deployed in cycle 144 (2026-09-19 13:39 UTC)
R-02	—	#107,064,266	Step 1 · native engine	1 change(s) of ours the network did not make	fixed: commit 12933ea, deployed in cycle 144 (2026-09-19 13:39 UTC)
R-03	—	#107,065,425	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-04	—	#107,067,322	Step 1 · native engine	15 change(s) of ours the network did not make	no disposition recorded
R-05	—	#107,070,519	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-06	—	#107,073,228	Step 1 · native engine	4 change(s) of ours the network did not make	no disposition recorded
R-07	—	#107,074,173	Step 1 · native engine	1 object(s) differed in content	fixed: commit 8f9636c, deployed in cycle 144 (2026-09-19 13:39 UTC)
R-08	—	#107,075,593	Step 1 · native engine	1 object(s) differed in content	no disposition recorded
R-09	—	#107,077,684	Step 1 · native engine	1 transaction result(s) differed from rippled's; 1 object(s) differed in content	no disposition recorded
R-10	—	#107,078,027	Step 1 · native engine	1 transaction result(s) differed from rippled's; 16 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: commit 949ed82, deployed in cycle 144 (2026-09-19 13:39 UTC)
R-11	—	#107,078,041	Step 1 · native engine	1 transaction result(s) differed from rippled's; 16 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-12	—	#107,078,052	Step 1 · native engine	1 transaction result(s) differed from rippled's; 16 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-13	—	#107,078,068	Step 1 · native engine	1 transaction result(s) differed from rippled's; 16 change(s) of ours the network did not make; 1 object(s) differed in content	no disposition recorded
R-14	—	#107,080,701	Step 1 · native engine	3 change(s) of ours the network did not make; 2 object(s) differed in content	fixed: commit cf2daea, deployed in cycle 144 (2026-09-19 13:39 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-18-01, INC-01 is INC-2026-09-18-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
141	`a1aecb0`	05:05:49 UTC	n/a	n/a	402 matched, 0 mismatched	#17 from 05:05 UTC: ended early: superseded by the next deployment (2026-09-18 16:11 UTC)
143	`312f235`	16:11:08 UTC	n/a	n/a	409 matched, 0 mismatched	#18 from 16:11 UTC: ended early: superseded by the next deployment (2026-09-19 13:39 UTC)

Cycle 141 delivered: `a1aecb0` fix(ledger): a multi-path pool is gated by its own book's tip, not only the rival strand (finding 298)

Cycle 143 delivered: `d4b2420` fix(node): Batch inner pairing is mode-aware for discarded all-or-nothing batches; unencodable inner ids keep their slot and trip a wire; `fda4f5f` feat(node): differential transaction fuzzer — unsigned mutants of real transactions, libxrpl vs our engine on the same pre-state; `7ee6c4d` feat(node): fuzzer — NOOP class for threading-only set differences, party roots in preflight bundles, mutation-diff readout; `a11f54c` fix(ledger): the fuzzer's first five findings — Payment shape rules, DeliverMin rules, IoC+FoK, sixteen-digit amounts (findings 299–303); `1581d6f` Merge branch 'tx-fuzz'; `78f33d9` feat(node): fuzzer — objects only our leg wrote become untouched pins in the bundle; `9ca58f4` test(ledger): a self-payment in one asset is temREDUNDANT, not a fee burn (finding 299); `01b5db2` feat(node): fuzzer — bundles carry our leg's bytes for every disputed key; `728f406` fix(node): fuzzer — the no-op test takes a pre-image from the native state when libxrpl never read the object; `312f235` Merge branch 'tx-fuzz'

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
13:23 UTC	xrpl-validator	`d4b2420`	fix(node): Batch inner pairing is mode-aware for discarded all-or-nothing batches; unencodable inner ids keep their slot and trip a wire
14:35 UTC	xrpl-validator	`fda4f5f`	feat(node): differential transaction fuzzer — unsigned mutants of real transactions, libxrpl vs our engine on the same pre-state
14:35 UTC	xrpl-validator	`7ee6c4d`	feat(node): fuzzer — NOOP class for threading-only set differences, party roots in preflight bundles, mutation-diff readout

Merged	Repo	Commit	Subject
14:35 UTC	xrpl-validator	`a11f54c`	fix(ledger): the fuzzer's first five findings — Payment shape rules, DeliverMin rules, IoC+FoK, sixteen-digit amounts (findings 299–303)
14:35 UTC	xrpl-validator	`1581d6f`	Merge branch 'tx-fuzz'
14:58 UTC	xrpl-validator	`9ca58f4`	test(ledger): a self-payment in one asset is temREDUNDANT, not a fee burn (finding 299)
15:56 UTC	xrpl-validator	`78f33d9`	feat(node): fuzzer — objects only our leg wrote become untouched pins in the bundle
15:56 UTC	xrpl-validator	`01b5db2`	feat(node): fuzzer — bundles carry our leg's bytes for every disputed key
15:56 UTC	xrpl-validator	`728f406`	fix(node): fuzzer — the no-op test takes a pre-image from the native state when libxrpl never read the object
15:56 UTC	xrpl-validator	`312f235`	Merge branch 'tx-fuzz'
16:12 UTC	xrpl-validator	`fdd63d7`	fix(node): fuzzer — the mutant carries its own transaction id before our leg applies it
16:12 UTC	xrpl-validator	`8c7e0a1`	Merge branch 'tx-fuzz'
16:15 UTC	xrpl-validator	`d8c566a`	fix(ledger): temRIPPLE_EMPTY ahead of every payment branch, the plain IOU send included (finding 300, placement)
16:24 UTC	xrpl-validator	`5872bed`	fix(ledger): a limit quality below the STAmount floor is rate 0 — the best quality, every strand rejected (finding 304)
16:56 UTC	xrpl-validator	`a816572`	fix(ledger): the post-crossing residual rests only while the account is still funded — read the account, not the clamped remainder (finding 305)
16:56 UTC	xrpl-validator	`4eba9b8`	Merge f305: post-crossing residual rests only while the account is still funded (finding 305)
17:31 UTC	xrpl-validator	`3d54768`	fix(ledger): a bridge round's direct strand whose pool is refused by its own tip is dead, not spared (finding 306)
17:31 UTC	xrpl-validator	`b8a1e20`	Merge f306-dead-strand: a direct strand refused by its own tip is dead, not spared (finding 306)
17:48 UTC	xrpl-validator	`85f37f9`	Merge f307-limitout-tiny: limitOut within 1e-9 is not a trim (finding 307)
17:48 UTC	xrpl-validator	`9220f94`	fix(ledger): limitOut hands the remainder back untrimmed within 1e-9 — no trim, no forgiveness (finding 307)
18:16 UTC	xrpl-validator	`8f9636c`	fix(ledger): a created PermissionedDomain files Flags 0 (finding 308)
18:16 UTC	xrpl-validator	`67b0d6d`	fix(ledger): accepting an IOU-priced NFT buy offer needs the buyer funded (finding 309)
18:16 UTC	xrpl-validator	`057bcd1`	fix(node): reset every per-transaction thread-local at the apply entry
18:16 UTC	xrpl-validator	`d4f329a`	Merge f308-domain-flags-nft-funds: domain Flags (F308), NFT buyer funds (F309), per-tx thread-state reset
18:16 UTC	xrpl-validator	`1061855`	feat(fuzz): bundles carry every base object our own walk read (sandbox read log)
18:16 UTC	xrpl-validator	`6fedc1c`	Merge fuzz-native-reads: fuzz bundles carry every base object our own walk read
18:53 UTC	xrpl-validator	`354efcd`	Merge f310-buy-out-fold: buy residual multiset fold (F310), pool-only overshoot flush (F311)
18:53 UTC	xrpl-validator	`12933ea`	fix(ledger): the carry overshoot flushes only through a pool the hop actually took (finding 311)

Merged	Repo	Commit	Subject
18:53 UTC	xrpl-validator	`624168c`	fix(ledger): a buy's residual folds the iterations' outs in multiset order (finding 310)
19:15 UTC	xrpl-validator	`e963a02`	fix(probe): hydrate every party's owner-directory root and tail, whatever the type
19:15 UTC	xrpl-validator	`7896337`	Merge dp-party-roots: the probe hydrates every party's owner-directory root and tail
19:25 UTC	xrpl-validator	`4cb253a`	Merge fuzz-structural-mutants: structural mutation kinds for the differential fuzzer
19:25 UTC	xrpl-validator	`35f4975`	feat(fuzz): structural mutation kinds — sequence gates, flags, modes, expirations
20:44 UTC	xrpl-validator	`fa8e282`	Merge f312-sequence-gates: sequence/fee gates (F312-F313), preflight shape rules (F314-F329), probe key hydration
20:44 UTC	xrpl-validator	`7e34f5f`	fix(probe): hydrate every 32-byte hex field of a transaction as a ledger key
20:44 UTC	xrpl-validator	`405b3f0`	fix(ledger): preflight and preclaim shape rules from the structural fuzzer and the testnet campaign (findings 314-329)
20:44 UTC	xrpl-validator	`f8341d6`	fix(ledger): the sequence, ticket, prior-transaction and last-ledger gates, and a zero fee is valid (findings 312, 313)
21:11 UTC	xrpl-validator	`a3c9afc`	feat(ledger): DepositPreauth by credentials — the credential-keyed entry (finding 319)
21:11 UTC	xrpl-validator	`da94ae3`	Merge f319-preauth-credentials: DepositPreauth by credentials (finding 319)
22:01 UTC	xrpl-validator	`c17d1db`	Merge f330-writeback-hex-case: the write-back test compares as rippled does (finding 330)
22:01 UTC	xrpl-validator	`feb7c23`	fix(ledger): the write-back test compares as rippled does — hex case-blind, nested, threading fields included (finding 330)
22:17 UTC	xrpl-validator	`de000dd`	fix(ledger): a Batch's outer fee level is not a preflight matter (finding 331)
22:17 UTC	xrpl-validator	`a295b93`	Merge f331-batch-fee-level: a Batch's outer fee level is not a preflight matter (finding 331)
22:19 UTC	xrpl-validator	`c2377fe`	Merge dp-party-roots-2: probe hydrates party roots, nested credential parties and the signer list
22:19 UTC	xrpl-validator	`74f9591`	fix(probe): hydrate every party's root, nested credential issuers and subjects, and the sender's signer list
22:34 UTC	xrpl-validator	`b29a764`	feat(ledger): hybrid domain offers rest in both books and leave both on deletion (finding 334)
22:34 UTC	xrpl-validator	`192ec17`	Merge f334-hybrid-offers: hybrid domain offers in both books (finding 334)
22:48 UTC	xrpl-validator	`546e3d8`	fix(ledger): a write-back equal to the base object is not a modification (finding 335)
22:48 UTC	xrpl-validator	`b83239b`	Merge f335-unchanged-writes: an unchanged write-back is not a modification (finding 335)
23:28 UTC	xrpl-validator	`949ed82`	Finding 336: FillOrKill sell fills only when the gross in-fold equals the budget exactly
23:28 UTC	xrpl-validator	`4671ddb`	fetch_tx_bundle: tolerate MPT amounts in the line, pair and pool sweeps
23:59 UTC	xrpl-validator	`0c589d7`	Finding 332: MPT token escrow — create locks the holder's token, finish/cancel unlock it
00:23 UTC	xrpl-validator	`3f67fa0`	Finding 333: DomainID payments and offers — accountInDomain gates and domain-book routing

Merged	Repo	Commit	Subject
00:57 UTC	xrpl-validator	`be0da22`	Finding 337: a domain offer whose owner left the domain is removed, not crossed
01:27 UTC	xrpl-validator	`c646018`	Finding 338: MPT escrow fee floors the delivery under fixCleanup3_4_0
01:39 UTC	xrpl-validator	`81c890a`	harness: hydrate an MPT escrow's issuance and party tokens for Finish/Cancel
02:11 UTC	xrpl-validator	`cf2daea`	Finding 339: a pool turn rejected by limitQuality ends the crossing

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(node): Batch inner pairing is mode-aware for discarded all-or-nothing batches; unencodable inner ids keep their slot and trip a wire** (`d4b2420`, xrpl-validator) Two rulings parked at the end of Batch leg B, pulled forward now that BatchV1_1 has majority on mainnet.
- **feat(node): differential transaction fuzzer — unsigned mutants of real transactions, libxrpl vs our engine on the same pre-state** (`fda4f5f`, xrpl-validator) `differential_probe --fuzz K [--seed S] [--fuzz-out DIR]`: for every transaction of a fixture ledger, before the real one is applied, K mutants (fuzz_mutants: amount ulp/ppm/halve/double nudges on the type's amount fields, Payment and OfferCreate flag toggles, DeliverMin set/drop, Paths drop, self-destination, Expiration past/future, OfferSequence drop, Fee double) are applied by libxrpl through the shim and by our …
- **feat(node): fuzzer — NOOP class for threading-only set differences, party roots in preflight bundles, mutation-diff readout** (`7ee6c4d`, xrpl-validator) Three lessons from the first 4,000 mutants. A preflight rejection reads nothing, so its bundle replayed natively as tefNO_ACCOUNT: bundles now seed `pre` with the sender's, destination's, owner's and issuer's roots as they stand. A set difference made only of Modified entries whose content equals the pre-image once PreviousTxnID/PreviousTxnLgrSeq are set aside is a threading-only write — rippled files a ModifiedNode …
- **fix(ledger): the fuzzer's first five findings — Payment shape rules, DeliverMin rules, IoC+FoK, sixteen-digit amounts (findings 299–303)** (`a11f54c`, xrpl-validator) From `differential_probe --fuzz` over #107052630, #107060755, #107009438 (4,204 unsigned mutants of real transactions, libxrpl 3.4.0 as the oracle): 310 of the 311 TER disagreements fall in five rules a validated ledger can never show the shadow, because a tem never reaches one.
- **feat(node): fuzzer — objects only our leg wrote become untouched pins in the bundle** (`78f33d9`, xrpl-validator) The probe then names the object when a replay repeats the divergence.
- **feat(node): fuzzer — bundles carry our leg's bytes for every disputed key** (`01b5db2`, xrpl-validator)
- **fix(node): fuzzer — the no-op test takes a pre-image from the native state when libxrpl never read the object** (`728f406`, xrpl-validator) A path's intermediate line written back byte-identical (park and take-back) is a NOOP, not a MUT.
- **fix(node): fuzzer — the mutant carries its own transaction id before our leg applies it** (`fdd63d7`, xrpl-validator) stamp_account_txn_id reads the id from the JSON hash; without it an armed AccountTxnID kept its old

value while libxrpl stamped the mutant's.

- **fix(ledger): temRIPPLE_EMPTY ahead of every payment branch, the plain IOU send included (finding 300, placement)** (`d8c566a`, xrpl-validator) Batch 7 still showed four pathless tfNoRippleDirect sends answering tesSUCCESS: the check sat inside the path-payment branch, which a pathless IOU send never enters.
- **fix(ledger): a limit quality below the STAmount floor is rate 0 — the best quality, every strand rejected (finding 304)** (`5872bed`, xrpl-validator) From the differential fuzzer: tfLimitQuality on a deliver-max partial payment (Amount 9999999999999990e79 RLUSD for 200000 drops). rippled's limit is Quality{Amounts{maxSourceAmount, dstAmount}} = getRate(Amount, SendMax); the quotient's exponent sits far below STAmount's floor of -96, `divide` canonicalizes it to zero, getRate files rate 0, and Quality(0) is the best quality there is — every strand's realised …
- **fix(ledger): the post-crossing residual rests only while the account is still funded — read the account, not the clamped remainder (finding 305)** (`a816572`, xrpl-validator) rippled re-reads accountFunds AFTER crossing (flowCross :438-446) and clears both sides of the residual when it is <= 0; the clamped remainder is not consulted. Fuzz #54 off 107060755 (B90D5364B3D9, TakerGets doubled) folded 25 fills one ulp short of the entry and rested an unfundable Offer plus its book page; fuzz #43 (5DA2F55D5276) spent its budget to exactly zero while the line kept 1e-17 and we rested nothing …
- **fix(ledger): a bridge round's direct strand whose pool is refused by its own tip is dead, not spared (finding 306)** (`3d54768`, xrpl-validator) rippled pushes only a strand that flowed back into the next iteration's active set (StrandFlow.h); a strand that ran and produced nothing is gone. Finding 215 modelled that but finding 211 spared the direct strand when its pool looked refused only by the rival bridge's bound. Fuzz #20 off 107009438 (7665065807E8, TakerGets doubled — r3rhWeE3's FLR→BTC offer over a direct pool and an XRP bridge): at iteration 8 the …
- **fix(ledger): limitOut hands the remainder back untrimmed within 1e-9 — no trim, no forgiveness (finding 307)** (`9220f94`, xrpl-validator) rippled's limitOut (StrandFlow.h:415-418): "A tiny difference could be due to the round off" — when the solved out is within 1e-9 relative of remainingOut (withinRelativeDistance on amounts, (max − min)/max < dist), remainingOut is returned unchanged, adjustedRemOut stays false, and the 1e-7 judge forgiveness at :731-733 never applies. Fuzz #121 off 107009438 (2249B3B51059 with tfLimitQuality added, rogue5Hn …
- **fix(ledger): a created PermissionedDomain files Flags 0 (finding 308)** (`8f9636c`, xrpl-validator) #107074173 06F51E33BB43 (soak #18): rippled's PermissionedDomainSet::doApply sets sfFlags to 0 on create, so the filed object is five bytes longer than ours — the Oracle lesson of finding 293 again. Vector from the receipt.
- **fix(ledger): accepting an IOU-priced NFT buy offer needs the buyer funded (finding 309)** (`67b0d6d`, xrpl-validator) #107063938 EBF238024D19 (soak #17): NFTokenAcceptOffer::preclaim reads the buyer's funds through accountFunds(view, owner, needed, fhZERO_IF_FROZEN) for every currency (NFTokenAcceptOffer.cpp:212-218). rDnNmaX1 held 0.3373 of the 337.002 xSPECTAR it offered: mainnet tecINSUFFICIENT_FUNDS, we accepted — nft_funds_short judged XRP prices only, its note that IOU prices are not settled being stale …
- **fix(node): reset every per-transaction thread-local at the apply entry** (`057bcd1`, xrpl-validator) rippled carries nothing from one transaction into the next. The offer walk's remembered owner counts, soft-stale and dead-reap marks, passthrough and self-maker credits, the pool walk's cells and context and Batch's inner collection are thread-locals that the PREVIOUS application leaves behind — and a discarded application (a differential-fuzz mutant, a dry run) leaves the most. Fuzz #85 off 107009438 reaped 21 …

- **feat(fuzz): bundles carry every base object our own walk read (sandbox read log)** (`1061855`, xrpl-validator) A fuzz bundle's pre-state was libxrpl's reads plus the party roots. Our walk can consult objects libxrpl never read — book pages it steps, lines it prices — and a bundle lacking them replays a narrower walk that happens to agree: fuzz #85 off 107009438 disagreed live (64 mutations vs 4) and passed from its bundle. The sandbox now keeps a thread-local read log (off by default; armed by the fuzzer around our leg) of …
- **fix(ledger): the carry overshoot flushes only through a pool the hop actually took (finding 311)** (`12933ea`, xrpl-validator) #107064266 12C8A416327C (soak-17 receipt): rogue5Hn's PLX→LHT→CSC payment met hop 1 with one book offer; the CSC/LHT pool merely existed, yet the 5.3e-11 LHT carry overshoot (finding for #106455266: a pool hop's excess rides through the pool) was flushed into it and its LHT line rounded up one ulp — a ninth mutation mainnet never wrote. rippled's fwd step feeds the excess through the stream it consumed, where an …
- **fix(ledger): a buy's residual folds the iterations' outs in multiset order (finding 310)** (`624168c`, xrpl-validator) rippled: afterCross.out -= result.actualAmountOut, and actualAmountOut is flow()'s sum(savedOuts) — the iterations' outs accumulated in the multiset's sorted order (StrandFlow.h). Finding 202 already kept that list in the bridged walker (saved_outs); the finding 144/160 tail then replaced it with the per-fill chain out_sum in consumption order. Fuzz #6 off 107009438 (24038BE2F358, TakerGets doubled — r3rhWeE3 buying …
- **fix(probe): hydrate every party's owner-directory root and tail, whatever the type** (`e963a02`, xrpl-validator) The per-type hydration blocks grew one specimen at a time; a transactor filing an owned object for a type without a block found no root in the state and native "created" a root that exists — #107074173 06F51E33BB43 (PermissionedDomainSet, 29-page dir), testnet 20863507 D958776AB6D7 (MPTokenIssuanceCreate) and 20863509 0AEE09161265 (MPTokenAuthorize) in one afternoon, each reported as a false DIVERGE-MUT. Every …
- **feat(fuzz): structural mutation kinds — sequence gates, flags, modes, expirations** (`35f4975`, xrpl-validator) Beyond amount nudges and a few flag toggles, the generator now emits: for every type, Sequence ±1 (terPRE_SEQ / tefPAST_SEQ), LastLedgerSequence behind the ledger (tefMAX_LEDGER) and a zero fee; TrustSet's fAuth, NoRipple, Freeze and DeepFreeze set/clear toggles, a zero limit (Nudge::Zero) and QualityIn/Out; AccountSet's asf set/clear values, TransferRate and TickSize edges; AMMDeposit/AMMWithdraw's exclusive mode …
- **fix(probe): hydrate every 32-byte hex field of a transaction as a ledger key** (`7e34f5f`, xrpl-validator) A modification that leaves an object byte-identical never reaches the meta, and a state without it reads tecNO_ENTRY: testnet 20863999 0DF65AD75341, a PermissionedDomainSet re-filing the same credential list, reported as a false DIVERGE. DomainID, CheckID, Channel, offer ids and credential ids now hydrate for every type.
- **fix(ledger): preflight and preclaim shape rules from the structural fuzzer and the testnet campaign (findings 314-329)** (`405b3f0`, xrpl-validator) F314 AccountSet: asfAuthorizedNFTokenMinter ⇔ NFTokenMinter (temMALFORMED); asfDisableMaster / asfNoFreeze need the master signature (tecNEED_MASTER_KEY, sigWithMaster = SigningPubKey deriving to the account); DisableMaster without RegularKey or SignerList is tecNO_ALTERNATIVE_KEY. F315 OfferCancel: account Sequence <= OfferSequence is temBAD_SEQUENCE. F316 TrustSet: tfSetfAuth without lsfRequireAuth is …
- **fix(ledger): the sequence, ticket, prior-transaction and last-ledger gates, and a zero fee is valid (findings 312, 313)** (`f8341d6`, xrpl-validator) rippled's Transactor::preclaim runs checkSeqProxy and checkPriorTxAndLastLedger before any transactor's own preclaim: a future Sequence is terPRE_SEQ, a used one tefPAST_SEQ, a TicketSequence not yet created terPRE_TICKET and one already consumed tefNO_TICKET, a

mismatched AccountTxnID tefWRONG_PRIOR, a LastLedgerSequence behind the ledger being built tefMAX_LEDGER. We incremented the account's Sequence without ever …

- **feat(ledger): DepositPreauth by credentials — the credential-keyed entry (finding 319)** (`a3c9afc`, xrpl-validator) XLS-70: DepositPreauth with AuthorizeCredentials files a DepositPreauth entry keyed by keylet::depositPreauth(owner, sorted credentials) — sha512Half('P' namespace, owner, [sha512Half(issuer, type)…], count), the count appended big-endian by beast::hash_append (verified against testnet 20864003's entry 61EEB051… by brute force over the variants). Preflight: exactly one of Authorize / Unauthorize / …
- **fix(ledger): the write-back test compares as rippled does — hex case-blind, nested, threading fields included (finding 330)** (`feb7c23`, xrpl-validator) ApplyStateTable skips a modified node only when *curNode* == origNode (ApplyStateTable.cpp:155), STObject equality over every field. Two ways our test disagreed with it: it compared hex strings case-sensitively and only at the top level (the transaction's lowercase hex against the decoder's uppercase — testnet 20863999 0DF65AD75341 re-filing the same credential list read as material and was threaded where rippled …
- **fix(ledger): a Batch's outer fee level is not a preflight matter (finding 331)** (`de000dd`, xrpl-validator) Batch::preflight checks that every inner fee is zero and nothing about the outer fee (Batch.cpp:326-334); the outer's LEVEL is Transactor::checkFee's, judged only while the ledger is open — finding 313's rule again. Devnet 5419046 1455FD412810 and 5419049 6A4D25E0FD9B, Fee 4 drops on a one-drop network, applied where we said temBAD_FEE. Verified by the full-ledger differential probe (a transaction bundle carries …
- **fix(probe): hydrate every party's root, nested credential issuers and subjects, and the sender's signer list** (`74f9591`, xrpl-validator) Three false DIVERGE-TERs from the testnet verification: CredentialCreate read tecNO_TARGET on a Subject root the state never held (20863992), DepositPreauth read tecNO_ISSUER on an issuer nested in AuthorizeCredentials (20864003), and an AccountSet asfDisableMaster read tecNO_ALTERNATIVE_KEY on a signer list that was never hydrated (20864007). native_read_keys now pushes each party field's account root, the roots …
- **feat(ledger): hybrid domain offers rest in both books and leave both on deletion (finding 334)** (`b29a764`, xrpl-validator) PermissionedDEX (live on mainnet): a tfHybrid domain offer is appended to the OPEN book at the same rate as well as the domain book — the open page carries no DomainID — records that entry in AdditionalBooks, and gains lsfHybrid (CreateOffer.cpp applyHybrid :536-580, :923-928); tfHybrid without a DomainID is temINVALID_FLAG (:76-77). Deleting the offer, by cancel or by crossing, unlinks every AdditionalBooks entry …
- **fix(ledger): a write-back equal to the base object is not a modification (finding 335)** (`546e3d8`, xrpl-validator) rippled's ApplyStateTable emits a modified node only when the working copy differs from the original (ApplyStateTable.cpp:155); a transactor that writes an object back unchanged has changed nothing. Testnet 20863999 0DF65AD75341 re-filed the same credential list on its domain: libxrpl's mutation set held the root alone, ours the root and the domain, a DIVERGE-MUT the threading fix of finding 330 could not close. …

Tests and campaigns

- **test(ledger): a self-payment in one asset is temREDUNDANT, not a fee burn (finding 299)** (`9ca58f4`, xrpl-validator)

Docs and operations

- **Merge branch 'tx-fuzz'** (`1581d6f`, xrpl-validator)

- **Merge branch 'tx-fuzz'** (`312f235` , xrpl-validator)
- **Merge branch 'tx-fuzz'** (`8c7e0a1` , xrpl-validator)
- **Merge f305: post-crossing residual rests only while the account is still funded (finding 305)** (`4eba9b8` , xrpl-validator)
- **Merge f306-dead-strand: a direct strand refused by its own tip is dead, not spared (finding 306)** (`b8a1e20` , xrpl-validator)
- **Merge f307-limitout-tiny: limitOut within 1e-9 is not a trim (finding 307)** (`85f37f9` , xrpl-validator)
- **Merge f308-domain-flags-nft-funds: domain Flags (F308), NFT buyer funds (F309), per-tx thread-state reset** (`d4f329a` , xrpl-validator)
- **Merge fuzz-native-reads: fuzz bundles carry every base object our own walk read** (`6fedc1c` , xrpl-validator)
- **Merge f310-buy-out-fold: buy residual multiset fold (F310), pool-only overshoot flush (F311)** (`354efcd` , xrpl-validator)
- **Merge dp-party-roots: the probe hydrates every party's owner-directory root and tail** (`7896337` , xrpl-validator)
- **Merge fuzz-structural-mutants: structural mutation kinds for the differential fuzzer** (`4cb253a` , xrpl-validator)
- **Merge f312-sequence-gates: sequence/fee gates (F312-F313), preflight shape rules (F314-F329), probe key hydration** (`fa8e282` , xrpl-validator)
- **Merge f319-preauth-credentials: DepositPreauth by credentials (finding 319)** (`da94ae3` , xrpl-validator)
- **Merge f330-writeback-hex-case: the write-back test compares as rippled does (finding 330)** (`c17d1db` , xrpl-validator)
- **Merge f331-batch-fee-level: a Batch's outer fee level is not a preflight matter (finding 331)** (`a295b93` , xrpl-validator)
- **Merge dp-party-roots-2: probe hydrates party roots, nested credential parties and the signer list** (`c2377fe` , xrpl-validator)
- **Merge f334-hybrid-offers: hybrid domain offers in both books (finding 334)** (`192ec17` , xrpl-validator)
- **Merge f335-unchanged-writes: an unchanged write-back is not a modification (finding 335)** (`b83239b` , xrpl-validator)
- **Finding 336: FillOrKill sell fills only when the gross in-fold equals the budget exactly** (`949ed82` , xrpl-validator)
- **fetch_tx_bundle: tolerate MPT amounts in the line, pair and pool sweeps** (`4671ddb` , xrpl-validator)
- **Finding 332: MPT token escrow — create locks the holder's token, finish/cancel unlock it** (`0c589d7` , xrpl-validator)
- **Finding 333: DomainID payments and offers — accountInDomain gates and domain-book routing** (`3f67fa0` , xrpl-validator)
- **Finding 337: a domain offer whose owner left the domain is removed, not crossed** (`be0da22` , xrpl-validator)
- **Finding 338: MPT escrow fee floors the delivery under fixCleanup3_4_0** (`c646018` , xrpl-validator)
- **harness: hydrate an MPT escrow's issuance and party tokens for Finish/Cancel** (`81c890a` , xrpl-validator)
- **Finding 339: a pool turn rejected by limitQuality ends the crossing** (`cf2daea` , xrpl-validator)

Independence program

No independence record for UTC day 2026-09-18; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
BatchV1_1	2026-09-15 14:06 UTC	2026-09-29 14:06 UTC

Amendments enabled on mainnet: 94.

Open items

- Receipt R-03 (#107,065,425): open, under investigation.
- Receipt R-04 (#107,067,322): open, under investigation.
- Receipt R-05 (#107,070,519): open, under investigation.
- Receipt R-06 (#107,073,228): open, under investigation.
- Receipt R-08 (#107,075,593): open, under investigation.
- Receipt R-09 (#107,077,684): open, under investigation.
- Receipt R-11 (#107,078,041): open, under investigation.
- Receipt R-12 (#107,078,052): open, under investigation.
- Receipt R-13 (#107,078,068): open, under investigation.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Validator Reputation Score Ledger Entry** (consensus) Add a new pseudo-transaction ttVALIDATOR_REPUTATION, applied only when included by a UNL supermajority at flag ledgers exactly like NegativeUNLModify,
- **Issuer Transfer Fee for MPT Securities** (rwa) Add sfTransferFee (uint16 basis points, 0-50000 matching the existing NFToken transfer-fee bound) and an lsfMPTCanTransferFee flag to MPTokenIssuance.
- **Expiring NFToken Usage Leases** (gaming) Add a new transaction NFTokenLease creating an owner-reserved ltNFTOKEN_LEASE object (keylet::nftLease(nftokenID)) carrying sfOwner, sfLesseeAccount,

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-18 13:23	09-18 13:23	xrpl-validator	`d4b2420`	fix(node): Batch inner pairing is mode-aware for discarded all-or-nothing batches; unencodable inner ids keep their slot and trip a wire
09-18 14:06	09-18 14:35	xrpl-validator	`fda4f5f`	feat(node): differential transaction fuzzer — unsigned mutants of real transactions, libxrpl vs our engine on the same pre-state
09-18 14:33	09-18 14:35	xrpl-validator	`7ee6c4d`	feat(node): fuzzer — NOOP class for threading-only set differences, party roots in preflight bundles, mutation-diff readout
09-18 14:35	09-18 14:35	xrpl-validator	`a11f54c`	fix(ledger): the fuzzer's first five findings — Payment shape rules, DeliverMin rules, IoC+FoK, sixteen-digit amounts (findings 299–303)
09-18 14:35	09-18 14:35	xrpl-validator	`1581d6f`	Merge branch 'tx-fuzz'
09-18 14:58	09-18 14:58	xrpl-validator	`9ca58f4`	test(ledger): a self-payment in one asset is temREDUNDANT, not a fee burn (finding 299)
09-18 14:48	09-18 15:56	xrpl-validator	`78f33d9`	feat(node): fuzzer — objects only our leg wrote become untouched pins in the bundle
09-18 15:12	09-18 15:56	xrpl-validator	`01b5db2`	feat(node): fuzzer — bundles carry our leg's bytes for every disputed key
09-18 15:45	09-18 15:56	xrpl-validator	`728f406`	fix(node): fuzzer — the no-op test takes a pre-image from the native state when libxrpl never read the object
09-18 15:56	09-18 15:56	xrpl-validator	`312f235`	Merge branch 'tx-fuzz'
09-18 16:12	09-18 16:12	xrpl-validator	`fdd63d7`	fix(node): fuzzer — the mutant carries its own transaction id before our leg applies it
09-18 16:12	09-18 16:12	xrpl-validator	`8c7e0a1`	Merge branch 'tx-fuzz'
09-18 16:15	09-18 16:15	xrpl-validator	`d8c566a`	fix(ledger): temRIPPLE_EMPTY ahead of every payment branch, the plain IOU send included (finding 300, placement)
09-18 16:24	09-18 16:24	xrpl-validator	`5872bed`	fix(ledger): a limit quality below the STAmount floor is rate 0 — the best quality, every strand rejected (finding 304)
09-18 16:52	09-18 16:56	xrpl-validator	`a816572`	fix(ledger): the post-crossing residual rests only while the account is still funded — read the account, not the clamped remainder (finding 305)
09-18 16:56	09-18 16:56	xrpl-validator	`4eba9b8`	Merge f305: post-crossing residual rests only while the account is still funded (finding 305)
09-18 17:28	09-18 17:31	xrpl-validator	`3d54768`	fix(ledger): a bridge round's direct strand whose pool is refused by its own tip is dead, not spared (finding 306)

Committed	Merged into main	Repo	Commit	Subject
09-18 17:31	09-18 17:31	xrpl-validator	`b8a1e20`	Merge f306-dead-strand: a direct strand refused by its own tip is dead, not spared (finding 306)
09-18 17:48	09-18 17:48	xrpl-validator	`85f37f9`	Merge f307-limitout-tiny: limitOut within 1e-9 is not a trim (finding 307)
09-18 17:48	09-18 17:48	xrpl-validator	`9220f94`	fix(ledger): limitOut hands the remainder back untrimmed within 1e-9 — no trim, no forgiveness (finding 307)
09-18 18:16	09-18 18:16	xrpl-validator	`8f9636c`	fix(ledger): a created PermissionedDomain files Flags 0 (finding 308)
09-18 18:16	09-18 18:16	xrpl-validator	`67b0d6d`	fix(ledger): accepting an IOU-priced NFT buy offer needs the buyer funded (finding 309)
09-18 18:16	09-18 18:16	xrpl-validator	`057bcd1`	fix(node): reset every per-transaction thread-local at the apply entry
09-18 18:16	09-18 18:16	xrpl-validator	`d4f329a`	Merge f308-domain-flags-nft-funds: domain Flags (F308), NFT buyer funds (F309), per-tx thread-state reset
09-18 17:50	09-18 18:16	xrpl-validator	`1061855`	feat(fuzz): bundles carry every base object our own walk read (sandbox read log)
09-18 18:16	09-18 18:16	xrpl-validator	`6fedc1c`	Merge fuzz-native-reads: fuzz bundles carry every base object our own walk read
09-18 18:53	09-18 18:53	xrpl-validator	`354efcd`	Merge f310-buy-out-fold: buy residual multiset fold (F310), pool-only overshoot flush (F311)
09-18 18:53	09-18 18:53	xrpl-validator	`12933ea`	fix(ledger): the carry overshoot flushes only through a pool the hop actually took (finding 311)
09-18 18:53	09-18 18:53	xrpl-validator	`624168c`	fix(ledger): a buy's residual folds the iterations' outs in multiset order (finding 310)
09-18 19:10	09-18 19:15	xrpl-validator	`e963a02`	fix(probe): hydrate every party's owner-directory root and tail, whatever the type
09-18 19:15	09-18 19:15	xrpl-validator	`7896337`	Merge dp-party-roots: the probe hydrates every party's owner-directory root and tail
09-18 19:25	09-18 19:25	xrpl-validator	`4cb253a`	Merge fuzz-structural-mutants: structural mutation kinds for the differential fuzzer
09-18 19:25	09-18 19:25	xrpl-validator	`35f4975`	feat(fuzz): structural mutation kinds — sequence gates, flags, modes, expirations
09-18 20:44	09-18 20:44	xrpl-validator	`fa8e282`	Merge f312-sequence-gates: sequence/fee gates (F312-F313), preflight shape rules (F314-F329), probe key hydration
09-18 20:44	09-18 20:44	xrpl-validator	`7e34f5f`	fix(probe): hydrate every 32-byte hex field of a transaction as a ledger key
09-18 20:44	09-18 20:44	xrpl-validator	`405b3f0`	fix(ledger): preflight and preclaim shape rules from the structural fuzzer and the testnet campaign (findings 314-329)
09-18 20:44	09-18 20:44	xrpl-validator	`f8341d6`	fix(ledger): the sequence, ticket, prior-transaction and last-ledger gates, and a zero fee is valid (findings 312, 313)
09-18 21:10	09-18 21:11	xrpl-validator	`a3c9afc`	feat(ledger): DepositPreauth by credentials — the credential-keyed entry (finding 319)
09-18 21:11	09-18 21:11	xrpl-validator	`da94ae3`	Merge f319-preauth-credentials: DepositPreauth by credentials (finding 319)
09-18 22:01	09-18 22:01	xrpl-validator	`c17d1db`	Merge f330-writeback-hex-case: the write-back test compares as rippled does (finding 330)
09-18 22:01	09-18 22:01	xrpl-validator	`feb7c23`	fix(ledger): the write-back test compares as rippled does — hex case-blind, nested, threading fields included (finding 330)
09-18 22:17	09-18 22:17	xrpl-validator	`de000dd`	fix(ledger): a Batch's outer fee level is not a preflight matter (finding 331)

Committed	Merged into main	Repo	Commit	Subject
09-18 22:17	09-18 22:17	xrpl-validator	`a295b93`	Merge f331-batch-fee-level: a Batch's outer fee level is not a preflight matter (finding 331)
09-18 22:19	09-18 22:19	xrpl-validator	`c2377fe`	Merge dp-party-roots-2: probe hydrates party roots, nested credential parties and the signer list
09-18 22:19	09-18 22:19	xrpl-validator	`74f9591`	fix(probe): hydrate every party's root, nested credential issuers and subjects, and the sender's signer list
09-18 22:34	09-18 22:34	xrpl-validator	`b29a764`	feat(ledger): hybrid domain offers rest in both books and leave both on deletion (finding 334)
09-18 22:34	09-18 22:34	xrpl-validator	`192ec17`	Merge f334-hybrid-offers: hybrid domain offers in both books (finding 334)
09-18 22:48	09-18 22:48	xrpl-validator	`546e3d8`	fix(ledger): a write-back equal to the base object is not a modification (finding 335)
09-18 22:48	09-18 22:48	xrpl-validator	`b83239b`	Merge f335-unchanged-writes: an unchanged write-back is not a modification (finding 335)
09-18 23:28	09-18 23:28	xrpl-validator	`949ed82`	Finding 336: FillOrKill sell fills only when the gross in-fold equals the budget exactly
09-18 23:28	09-18 23:28	xrpl-validator	`4671ddb`	fetch_tx_bundle: tolerate MPT amounts in the line, pair and pool sweeps
09-18 23:59	09-18 23:59	xrpl-validator	`0c589d7`	Finding 332: MPT token escrow — create locks the holder's token, finish/cancel unlock it
09-19 00:23	09-19 00:23	xrpl-validator	`3f67fa0`	Finding 333: DomainID payments and offers — accountInDomain gates and domain-book routing
09-19 00:57	09-19 00:57	xrpl-validator	`be0da22`	Finding 337: a domain offer whose owner left the domain is removed, not crossed
09-19 01:27	09-19 01:27	xrpl-validator	`c646018`	Finding 338: MPT escrow fee floors the delivery under fixCleanup3_4_0
09-19 01:39	09-19 01:39	xrpl-validator	`81c890a`	harness: hydrate an MPT escrow's issuance and party tokens for Finish/Cancel
09-19 02:11	09-19 02:11	xrpl-validator	`cf2daea`	Finding 339: a pool turn rejected by limitQuality ends the crossing

Times in UTC. A commit appears on the day it was committed to main or merged into it.